

Chapter 36: The Puerperium — Revision Notes

Williams Obstetrics 26e, pp. 1633–1672 of the PDF. High-yield numbers are in **bold**. The five tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 7 figures are included from the PDF.

Big picture: The puerperium (Latin *puer*, child + *parus*, bringing forth) is the **4–6 weeks** after delivery during which pregnancy changes reverse. Morbidity is surprisingly common: **3%** of British mothers were readmitted within 8 wk, and almost **¾** still had health problems up to 18 months later.

Table 36-1. Concerns Raised by Women in the Puerperium

Concerns	Percent
Pain after cesarean delivery	58
Feeling stressed	54
Breastfeeding issues	48
Perineal pain after vaginal delivery	41
Need for social support	32
Inadequate education about newborn care	21
Help with postpartum depression	10
Perceived need for extended hospital stay	8
Need for maternal insurance coverage postpartum	6

From *Childbirth Connection*, 2013a,b; Kanotra, 2007 (PRAMS data).

1. The Fourth Trimester

- ACOG (2018) concept: the **12 weeks after birth** are a critical period → comprehensive physical, social and psychological assessment.
- **Initial visit at 3 wk** postdelivery and a **final summary visit at 12 wk**; add visits as needed (chronic hypertension, overt diabetes, cardiovascular disease, depression).
- Use it to discuss lifestyle changes and **update immunizations**. Afterwards → well-woman care, primary/specialty care, preconceptional counseling.

Table 36-2. Components of Postpartum Care for the Fourth Trimester

Components
Care team
Postpartum visits
Lactation support
Infant feeding plan
Reproductive life plan
Contraception
Pregnancy complications
Cardiovascular risk assessment
Mental health
Postpartum problems
Chronic conditions

2. Reproductive Tract Involution

Birth canal

- Vagina and outlet shrink but **rarely regain nulliparous dimensions. Rugae reappear by 3 wk** (less prominent).
- Hymenal remnants scar into **myrtiform caruncles**.
- Vaginal epithelium is hypoestrogenic; it doesn't proliferate until **4–6 wk** (= resumed ovarian estrogen).
- Parturition predisposes to pelvic organ prolapse.

Cervix

- For a few days it **admits two fingers**; by the **end of week 1** it narrows, thickens and the canal reforms.
- The external os stays wider, with permanent lateral depressions at laceration sites → **parous cervix** (transverse slit).
- Epithelial remodeling: **almost half of women have regression of high-grade dysplasia** after delivery.

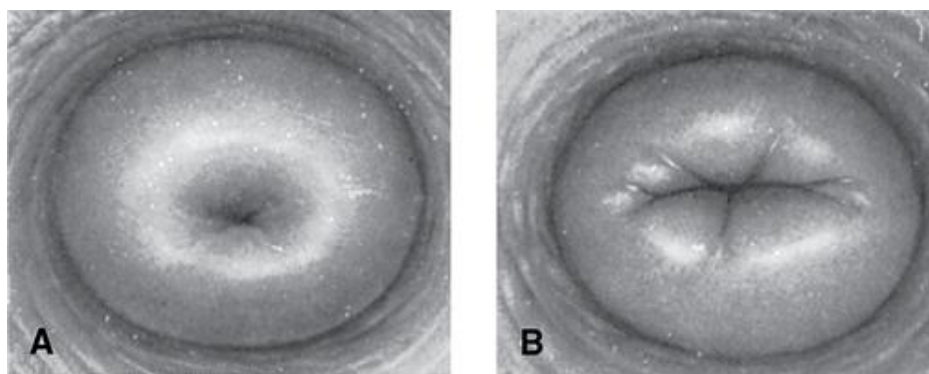


Figure 36-1. Nulliparous cervix (A, round os) vs parous cervix (B, transverse slit with laceration scars).

Uterus

- Pelvic vessels shrink; large uterine vessels are obliterated by **hyaline change** and replaced by smaller ones.
- Immediately postpartum: fundus **slightly below the umbilicus**; walls **4–5 cm** thick; the lower segment becomes the barely discernible **isthmus** over a few weeks.

Time postpartum	Uterine weight
Immediately	~1000 g
1 wk	~500 g
2 wk	~300 g
4 wk	~100 g (involution complete)

- Involution starts by **day 2**. Myocyte **number doesn't fall much — cell size falls**. Each pregnancy leaves the uterus slightly larger.

Sonographic findings

- Uterine size falls by more than half, **450 → 200 g, in the first 2 wk**. Uterus and endometrium return to pregravid size by **8 wk**. Multiparas involute more slowly.
- After cesarean, residual myometrium over the scar is thicker with **two-layer closure**.
- **Fluid in the cavity is normal**: 78% at 2 wk, 52% at 3 wk, 30% at 4 wk, 10% at 5 wk.
- **Intracavitary tissue** (3-D US): 1/3 on day 1, **95% on day 7**, 87% day 14, 28% day 28; empty by day 56.
- Uterine artery resistance rises steadily for 7 wk; no difference between vaginal and cesarean delivery.

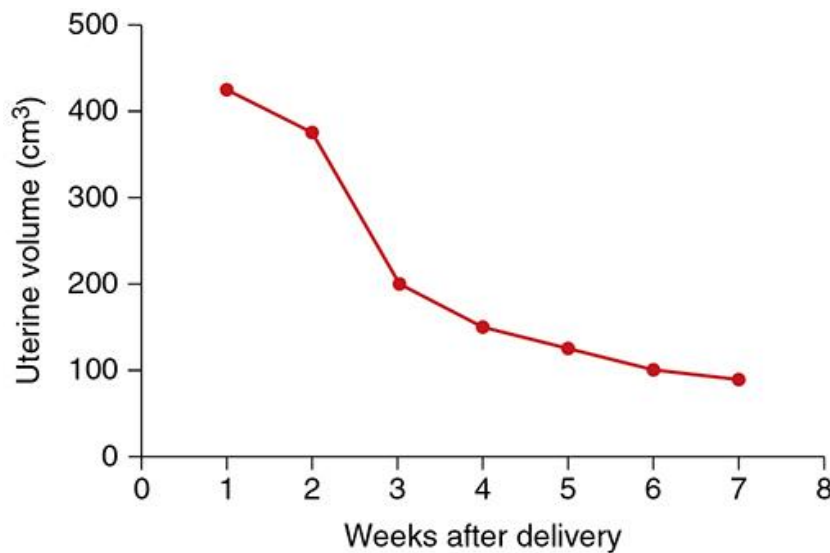


Figure 36-2. Sonographic uterine volume falls steeply in the first 3 wk (~425 → 200 cm³), then slowly to ~90 cm³ by 7 wk.

Decidual and endometrial regeneration

- Separation occurs in the **spongy layer**, so the **decidua basalis is not shed**.
- By **2–3 days** it splits into two layers: **superficial → necrotic → lochia**; **basal → source of new endometrium**.
- Surface re-epithelialized in **~1 wk**; **fully restored endometrium from day 16** (except the placental site).
- **Histological endometritis is normal** (repair). Noninfectious "salpingitis" changes in **~1/2** of women at 5–15 days.

Clinical aspects

Afterpains

- Primiparas: uterus stays tonically contracted. **Multiparas**: intermittent vigorous contractions.
- ↑ with parity and **worse with suckling (oxytocin)**; mild by **day 3**.
- Unusually severe/persistent afterpains → think **uterine infection**.

Lochia (RBCs, shredded decidua, epithelial cells, bacteria)

Type	Timing	Color / content
Lochia rubra	First 3–4 days	Red (blood)
Lochia serosa	After 3–4 days	Progressively paler
Lochia alba	After ~day 10	White/yellow-white (leukocytes, less fluid)

- Average total duration **24–36 days**.
- Because leukocytes are expected, **saline microscopy of lochia is uninformative** for suspected metritis.

3. Placental Site Involution

- Complete extrusion takes **up to 6 wk**. Palm-sized at delivery (full of thrombosed vessels) → **3–4 cm by end of week 2**.
- Mechanism = **exfoliation**, not absorption in situ: undermined by new endometrium growing in from the margins and up from glands deep in the decidua basalis; infarcted superficial tissue sloughs.

Subinvolution

- **Causes:** incompletely remodeled **spiral arteries**, retained placental fragments, infection.
 - Noninvolved vessels: thrombosed, no endothelium, perivascular trophoblast.
- **Features:** prolonged lochia, irregular/excessive bleeding; uterus **larger and softer** than expected.
- **Sonography:**
 - **Retained products** → thickened endometrium or endometrial mass; **vascularity ↑ the likelihood**.
 - **Subinvolution** → enlarged uterus with **tubular hypoechoic areas in the myometrium** (dilated vessels).
- **Treatment (Parkland):**
 - **Methylergonovine 0.2 mg PO q3–4h for 24–48 h** (efficacy of uterotonics is questionable).
 - Brisk bleeding: IM methylergonovine ± **oxytocin 20 U in 1 L** crystalloid.
 - Suspected infection: 1/3 of late metritis is ***Chlamydia trachomatis*** → oral, twice daily for **7–10 days**: **azithromycin 500 mg**, **amoxicillin-clavulanate 875 mg** or **doxycycline 100 mg**.

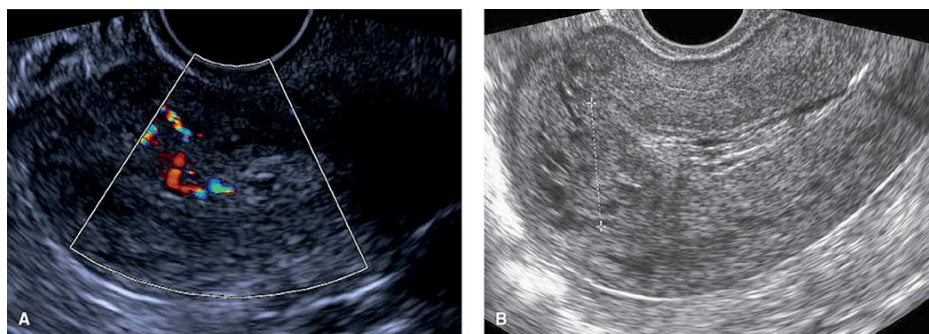


Figure 36-3. (A) Retained products with color Doppler vascularity; (B) retained clot (calipers) with dilated fundal myometrial vessels in subinvolution.

Late postpartum hemorrhage

- **Secondary PPH (ACOG) = bleeding 24 h to 12 wk after delivery.** Worrisome bleeding at 1–2 wk in **~1%**.
- **Causes:** most often **abnormal placental-site involution**; occasionally a retained fragment or **uterine artery pseudoaneurysm**; **von Willebrand disease** or other coagulopathies.
- **Placental polyp:** retained products → necrosis, fibrin, neovascularization. Severe bleeding in only **6%**. US: discrete hypervascular endometrial mass extending into myometrium.
- **Management:**
 - Few have retained fragments, so **no routine curettage** (it may avulse the implantation site and worsen bleeding).

- Stable + **empty cavity on US** → **oxytocin, methylergonovine or a prostaglandin** ± antimicrobials if infection suspected.
- **Large clots on US** → gentle suction curettage. Otherwise curette only if bleeding persists or recurs after medical therapy.

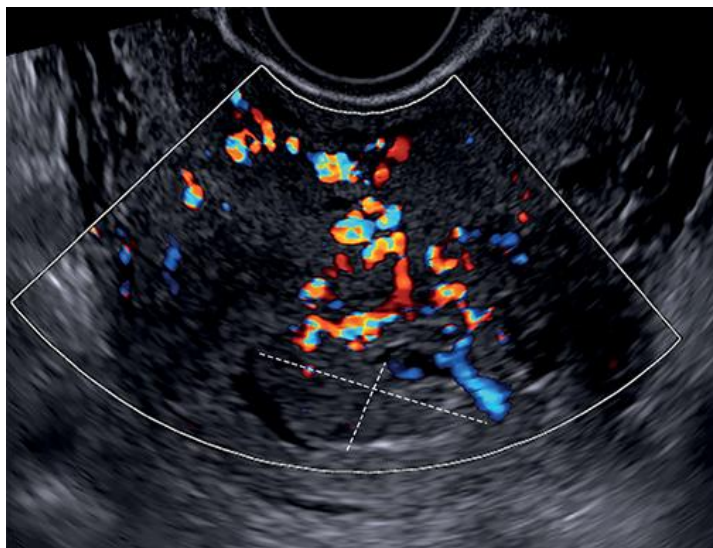


Figure 36-4. Placental polyp (dashed lines): hypervascular mass extending into the myometrium on color Doppler.

4. Urinary Tract

- Glomerular **hyperfiltration persists** but is back to baseline by **2 wk**.
- **Dilated ureters/pelves normalize by 2–8 wk** → UTI risk (plus residual urine, bacteriuria, bladder trauma).
- Bladder trauma (submucosal hemorrhage/edema) correlates with **labor length**.
- Postpartum bladder: **greater capacity, relative insensitivity** to pressure → overdistention, incomplete emptying.
- **Retention is more common with epidural or narcotic analgesia**.
- Puerperal **stress urinary incontinence ~5%**.

5. Peritoneum and Abdominal Wall

- Broad and round ligaments take considerable time to recover. Abdominal wall soft and flaccid (ruptured elastic fibers).
- Lax abdomen → ordinary girdle or binder. **Exercise:** any time after vaginal delivery; **wait ~6 wk after cesarean** (fascial healing).
- **Diastasis recti** = thinning/widening of the linea alba. **Musculofascial continuity and no true hernia sac** distinguish it from a ventral hernia.

6. Blood and Blood Volume

Hematological and coagulation changes

- **WBC seldom >25,000/ μ L** (mainly granulocytes); relative lymphopenia, absolute eosinopenia. Thrombocytosis possible.
- Hb/Hct fall moderately in the first days; a fall **much below prelabor** values = considerable blood loss.
- **Fibrinogen stays high for ≥ 1 wk** → hypercoagulability. **DVT/PE risk persists for 12 wk** after childbirth.

Pregnancy-induced hypervolemia

- If the pregnancy hypervolemia is lost as PPH → nonpregnant volume regained almost immediately. Otherwise blood volume returns to near nonpregnant by **1 wk**.

Parameter	Postpartum course
Cardiac output	Stays ↑ 24–48 h , nonpregnant by 10 days
Heart rate	Same pattern as CO
Blood pressure	Rises toward nonpregnant values over the first days
SVR	Stays low for 2 days , then rises steadily
Arterial stiffness	Reduction persists (favorable remodeling → may explain ↓ preeclampsia risk in later pregnancies)

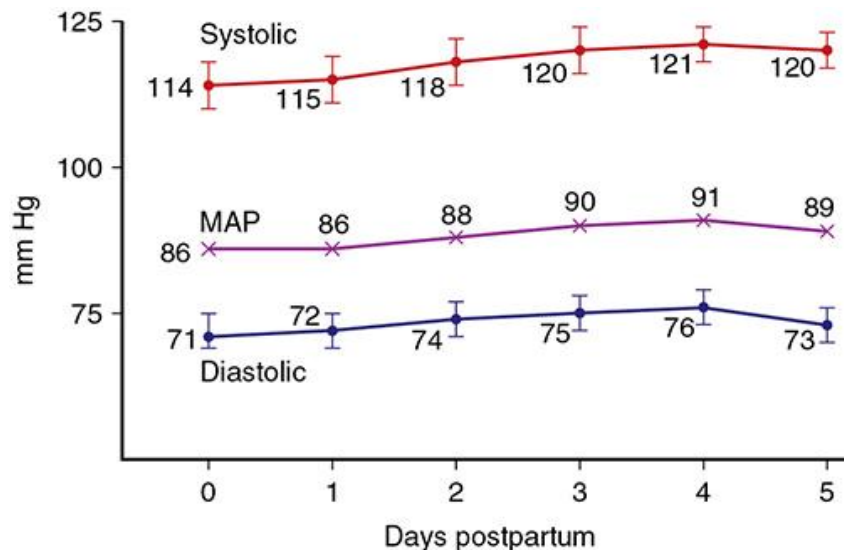


Figure 36-5. BP normally rises over days 0–5 postpartum (systolic 114 → ~120, diastolic 71 → ~75, MAP 86 → ~90 mm Hg).

Postpartum diuresis

- Physiological reversal of Na⁺/water retention: **sodium space ↓ ~2 L in week 1**. Can be **prodigious in preeclampsia**.
- Weight loss: **5–6 kg at delivery** (baby + blood loss) + **2–3 kg from diuresis**. Weight loss peaks by the end of **week 2**; residual excess = fat stores.

7. Lactation and Breastfeeding

Breast anatomy and secretory products

- **15–25 lobes** → lobules → alveoli. Each lobe has one lactiferous duct opening separately on the nipple.
- **Colostrum**: deep yellow, expressible by **day 2**.
 - vs mature milk: **more immune components, minerals, amino acids and protein (mostly globulin); less sugar and fat**.
 - Lasts **5–14 days**; mature milk by **4–6 wk**.
 - **IgA** protects against enteric pathogens. Also complement, macrophages, lymphocytes, lactoferrin, lactoperoxidase, lysozymes.
- **Milk output ≥600 mL/d**; gestational weight gain has little effect on quantity or quality.
- **Milk is isotonic with plasma; lactose = ½ the osmotic pressure**.
- Unique proteins: **α-lactalbumin, β-lactoglobulin, casein**. Fatty acids made in alveoli from glucose (apocrine-like secretion).

- **Vitamin K is virtually absent** → IM vitamin K to the newborn. **Vitamin D is low** → newborn supplementation.
- **Whey:casein = 60:40** (ideal for absorption). Whey is rich in IL-6.
- Others: prolactin, **epidermal growth factor** (resists gastric proteolysis → gut maturation), melatonin, oligosaccharides.

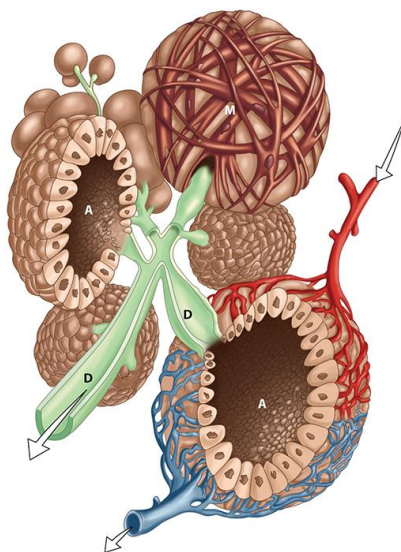


Figure 36-6. Alveolus (A) secretes milk; myoepithelial fibers (M) contract under oxytocin to eject it into ducts (D).

Table 36-3. Average Composition of Human Breast Milk

Component	g/100 mL
Fat — Total	4.2
Fatty acids	Trace
PUFA	0.6
Cholesterol	0.016
Protein — Total	1.1
Casein	0.3
α-Lactalbumin	0.3
Lactoferrin	0.2
Carbohydrate — Lactose	7
Oligosaccharides	0.5

PUFA = polyunsaturated fatty acids.

Lactation endocrinology

- Growth of the secretory apparatus: progesterone, estrogen, placental lactogen, prolactin, cortisol, insulin.
- **Delivery** → **abrupt fall in progesterone and estrogen**:
 - Removes progesterone's inhibition of **α-lactalbumin** → **lactose synthase** ↑ → milk lactose ↑.
 - Prolactin acts **unopposed**.
- Calcium-sensing receptors ↓ PTHrP and ↑ Ca transport into milk. Serotonin helps maintain production.
- **Maintenance = repeated suckling and emptying.**
- **Prolactin is essential** → **Sheehan syndrome = no lactation**. Baseline prolactin falls after delivery, but **each suckling episode causes a surge** (↓ hypothalamic **dopamine = prolactin-inhibiting factor**).
- **Oxytocin** (posterior pituitary, pulsatile) → myoepithelial contraction → **milk ejection ("let-down")**. Triggered by suckling or even an **infant's cry**; **inhibited by fright or stress**.

Memory hook: "Prolactin **P**roduces, Oxytocin **O**ut-puts (ejects)."

Immunological consequences

- Secretory IgA and growth factors; antibodies target maternal environmental antigens (e.g., *E. coli*).
- ↓ **ear, respiratory and GI infections, NEC and SIDS**. Especially important for preterm infants.
- Milk T cells are mostly **memory T cells** → the neonate benefits from maternal immune experience.

Lactation (benefits and uptake)

- **Start breastfeeding within 1 h of birth**. WHO/AAP/ACOG: **exclusive breastfeeding for up to 6 months**.
- **Maternal:** ↓ breast and reproductive cancer; less postpartum weight retention.
- **Child:** higher adult IQ scores; ↓ SIDS.
- A 90% breastfeeding rate for 12 months would save >\$3 billion/yr (US).
- Uptake: **55%** breastfeeding at 6 months (Healthy People 2020 goal **61%**); **<2%** exclusively breastfed at discharge; **<1/2** at 3 months.
- **Baby-Friendly Hospital Initiative:** ~20,000 hospitals worldwide; only **15–20% of US hospitals**.

Table 36-4. Advantages of Breastfeeding

Advantages
Nutritional
Immunological
Developmental
Psychological
Social
Economic
Environmental
Optimal growth and development
Decrease risks for acute and chronic diseases

Table 36-5. Ten Steps to Successful Breastfeeding

#	Step
1	Have a written breastfeeding policy that is regularly communicated to all health-care staff
2	Train all staff in skills necessary to implement this policy
3	Inform all pregnant women about the benefits and management of breastfeeding
4	Help mothers initiate breastfeeding within an hour of birth
5	Show mothers how to breastfeed and how to sustain lactation, even if they should be separated from their infants
6	Feed newborns nothing but breast milk, unless medically indicated, and prioritize donor breast milk when supplementation is needed
7	Practice rooming-in , which allows mothers and newborns to remain together 24 hours a day
8	Encourage breastfeeding on demand
9	Give no artificial pacifiers to breastfeeding newborns
10	Help start breastfeeding support groups and refer mothers to them

Adapted from the World Health Organization, 2018.

Breast care

- Cleanliness; wash areola with water and mild soap before and after feeds.
- **Fissured nipples:** painful, ↓ milk, portal for pyogenic bacteria → topical **lanolin + nipple shield for ≥24 h**. **Severe fissuring** → **don't nurse on that side; pump** until healed.
- Cause is usually **poor latch** (nipple only, pressed against the hard palate). Correct latch = **nipple + areola**.

Breastfeeding contraindications

Contraindicated	NOT contraindicated
Street drugs or uncontrolled alcohol use	CMV (virus and antibodies both in milk)
Infant with galactosemia	Hepatitis B — if newborn gets HBIG
HIV (in developed countries)	Hepatitis C (no proven transmission)
Active, untreated TB	HSV — if no breast lesions + hand washing
Certain medications	COVID-19 — mask + hand hygiene
Breast cancer treatment	

Drugs secreted in milk

- Most drugs appear in milk but infant doses are small. Determinants: plasma concentration, protein binding, pH, ionization, **lipid solubility**, molecular weight. **Milk-to-plasma ratio**.
- **Prefer:** shorter half-life, poor oral absorption, low lipid solubility.
- **Timing:** multiple daily doses → take **right after a feed**; once-daily → **before the longest infant sleep** (bedtime).
- **Cytotoxics** (cyclophosphamide, cyclosporine, doxorubicin, methotrexate, mycophenolate) are a concern. Avoid **marijuana and alcohol**. Resource: **LactMed**.
- **Radioisotopes** (Cu, Ga, In, I, Na, Tc) enter milk rapidly → pick the shortest-excreted agent; **pump and freeze beforehand**, then pump and discard for **15 h to 2 wk** depending on isotope. Radioactive iodine concentrates in the thyroid (Ch 66).
- **Gadolinium: don't interrupt breastfeeding.**

Breast engorgement

- Common in **non-breastfeeding** women: leakage and pain **peaking at 3–5 days**. Up to ½ need analgesia; **10%** severe pain up to 14 days.
- **Treatment:** supportive bra/binder, **cool packs, oral analgesics for 12–24 h**. **Pharmacological suppression of lactation is not recommended.**
- **Engorgement fever:** 13% of cases, **37.8–39°C**, seldom lasts **>4–16 h**. Exclude infection (mastitis, Ch 37).

Other lactation issues

- **Inverted nipples:** pump if shallow; if deep, "tease" the nipple out daily in late pregnancy.
- **Polymastia / polythelia** along the **milk line (axilla → groin)**; incidence **0.22–6%**. May enlarge or engorge; can mimic moles, lymph nodes or lipoma. No obstetric significance.
- **Galactocele:** duct blocked by inspissated milk → fluctuant mass mimicking an abscess; resolves or aspirate.
- Milk volume depends on **glandular development**, not general health. **Agalactia** (none) and **polygalactia** (excess) are rare.
- **Lactation-associated osteoporosis:** rare; back pain, vertebral fractures; **bisphosphonates** seem effective. Lactation does **not** cause later osteoporosis in almost all women.

8. Hospital Care

Immediate monitoring

- **BP and pulse every 15 min for ≥2 h. Temperature every 4 h for 8 h**, then at least every 8 h.
- After regional or general anesthesia → staffed recovery area.
- Watch vaginal bleeding and **palpate the fundus**; if relaxed → **massage** ± uterotonics. Concealed intrauterine blood → uterine enlargement.
- **Early ambulation** (attendant present the first time): fewer bladder problems, less constipation, ↓ **VTE** (Parkland attributes its very low VTE rate to this).
- **Diet**: eat **2 h** after uncomplicated vaginal delivery. Lactation → slightly more calories and protein.
- **Oral iron for ≥3 months** postpartum; check Hct at the first visit.
- **Hot flushes** (estrogen withdrawal), especially at night → check temperature to exclude infection.
- **Headache**: migraine may flare with hypoestrogenism; exclude **postdural puncture headache** and **hypertensive complications**. Ibuprofen/acetaminophen → narcotics; **sumatriptan is breastfeeding compatible**.

Perineal care

- Clean front to back. **Cool pack in the first 24 h**; local anesthetic spray; **warm sitz baths from ~24 h**. Tub bathing allowed.
- **Severe perineal/vaginal/rectal pain always needs examination: hematoma in the first day or so; infection after day 3–4.**
- Episiotomy healed and nearly asymptomatic by **3 wk**.
- **Puerperal procidentia**: improves with involution; temporize with a **donut pessary**.
- **Thrombosed hemorrhoids** (worsened by pushing): topical anesthetics, warm soaks, stool softeners. OTC steroid/astringent/phenylephrine preparations have no RCT support.

Bladder function

- Retention is common: IV fluids + **antidiuretic effect of oxytocin** + ↓ sensation. **~5%** on bladder scan (much less clinically).
- **Risk factors: primiparity, epidural, cesarean, perineal lacerations, operative vaginal delivery, catheterization in labor, prolonged 2nd stage.**
- Detect by suprapubic palpation or a fundus pushed **above the umbilicus**; most units use automated bladder sonography.

Management algorithm

1. **Not voided within 4 h** of delivery → she probably can't. Examine for **perineal/genital tract hematoma**.
2. Overdistended bladder → **indwelling catheter for ≥24 h** (prevents recurrence, restores tone).
3. Remove → **voiding trial**. Can't void by 4 h → scan: **>200 mL → re-catheterize for another 24 h; <200 mL → remove catheter** and monitor.
4. Failed second trial (rare) → indwelling catheter + leg bag; outpatient voiding trial in **1 wk**. All protracted cases resolved by **3 wk**.
5. **40% develop bacteriuria** → single dose or short course of antibiotics after removal.

Pain, mood and cognition

- Analgesia: codeine, ibuprofen, acetaminophen (combinations), up to **every 4 h**.
- **Screen for depression.**
- **Postpartum blues**: emotional letdown, discomfort, sleep deprivation, anxiety, body image. **Mild, self-limited 2–3 days (up to 10 days)**. Treat with anticipation, recognition, reassurance. **Persistent/worse → evaluate for major depression.**
- Executive function may decline postpartum.

Neuromusculoskeletal problems

Obstetrical neuropathies (~1% of puerperas)

- Mechanism: lumbosacral plexus compressed by the fetal head or forceps at the pelvic brim; **common fibular nerve compressed in stirrups**.
- **Footdrop** → lumbosacral plexus, sciatic or common fibular (peroneal) nerve injury.
- Most common: **lateral femoral cutaneous (24%)**, then **femoral (14%)**. Motor deficit in $\frac{1}{3}$.
- **Risk factors**: nulliparity, prolonged 2nd stage, long pushing in **semi-Fowler position**.
- **Median duration 2 months** (2 wk–18 months).
- Cesarean → iliohypogastric/ilioinguinal nerve injury.

Musculoskeletal injuries

- Pelvic girdle/hip pain from stretching or tearing → **MRI** if exam normal; NSAIDs + physical therapy. Rare **septic pyomyositis** (iliopsoas abscess).
- **Symphyseal separation**: 1 in 600 to 1 in 30,000. Pain onset acute at delivery (can be antepartum or up to 48 h after).
 - **Normal symphysis 0.4–0.5 cm; >1 cm = diastasis** (radiograph).
 - Treat conservatively: **lateral decubitus rest + pelvic binder**; surgery occasionally if **>4 cm**.
 - **High recurrence** → consider cesarean next time.
- Rare: sacral or pubic ramus fracture (more likely with heparin/steroid osteoporosis); **osteitis pubis** (osteomyelitis).



Figure 36-7. Pubic symphyseal separation on postpartum day 1: shuffling gait, treated with a pelvic binder, analgesia and physical therapy.

Immunizations

- **D-negative, unsensitized, D-positive newborn** → **300 µg anti-D immune globulin** shortly after delivery.
- **Rubella and varicella** vaccines before discharge if nonimmune.
- **Tdap/Td and influenza** if not given; **COVID-19 vaccine** (including breastfeeding women).
- EMR best-practice alert → **97%** Tdap rate at Parkland. ACOG supports **standing orders**.

Contraception

- Immediate puerperium is ideal for **LARC**.
- **Non-breastfeeding**: menses return in **6–8 wk**; **ovulation at a mean of 7 wk (5–11 wk)**, but **before 28 days** has been described → conception possible early.
- 58% of adolescents were sexually active by 3 months; only 80% of them used contraception.

- **Breastfeeding:** ovulation much less frequent but variable; first menses **2nd–18th month**. Campbell and Gray findings:
 1. Resumed ovulation is often marked by normal menstrual bleeding.
 2. **Feeds of 15 min, 7× daily** delayed ovulation.
 3. Ovulation can occur **without bleeding**.
 4. Bleeding can be **anovulatory**.
 5. Pregnancy risk with breastfeeding **~4%/yr**.
- **Progestin-only methods** (pills, DMPA, implant, LNG-IUD, progesterone vaginal ring) **don't affect milk** → start any time.
- **Estrogen-progestin methods probably ↓ milk quantity**, but can be used in the right circumstances.

Hospital discharge

- **Stay up to 48 h after uncomplicated vaginal delivery; up to 96 h after cesarean** (federal law). Earlier discharge is fine if desired and appropriate.
- Teach: lochia, diuretic weight loss, let-down. **Warning signs:** fever, heavy bleeding, leg pain/swelling/tenderness. **Persistent headache, dyspnea or chest pain → immediate concern.**

9. Home Care

Coitus

- No evidence-based rule; **after 2 wk** based on desire and comfort. 60% resume by 1 wk, 80% by 4 months; 1/3 have sexual dysfunction.
- **Dyspareunia:** only **0.4%** with 1st/2nd-degree tears (no episiotomy) vs primiparas with episiotomy: **67% dysfunction at 3 months, 31% at 6, 15% at 12**. Only **40%** with anal sphincter injury resumed by 12 wk. Also common after cesarean.
- **Hypoestrogenic thin epithelium** (prolonged with breastfeeding) → poor lubrication and dysuria → **topical estrogen cream** daily for several weeks + lubricants (exclude cystitis first).

Follow-up care

- Most activities (bathing, driving, housework) resume by discharge.
- Only **1/2** regain usual energy by 6 wk; **vaginal delivery 2× more likely** than cesarean to have normal energy.
- **80%** of working women return to work within 1 yr.
- ACOG: **comprehensive visit within 12 wk** after delivery.

Quick-Recall: Postpartum Timeline

When	What happens
First 2 h	BP/pulse q15 min ; fundal checks; eat at 2 h
4 h	Not voided → probably can't → catheterize
24 h	Switch from cool packs to warm sitz baths ; secondary PPH window starts
24–48 h	Cardiac output still ↑; SVR still low (2 days)
Day 2	Myometrial involution starts; colostrum expressible
Days 2–3	Afterpains mild by day 3; decidua splits into 2 layers; blues last 2–3 days
Days 3–4	Lochia rubra → serosa ; perineal pain now suggests infection
Days 3–5	Engorgement peaks
Day 10	Lochia alba ; cardiac output nonpregnant
1 wk	Uterus 500 g ; cervix reformed; blood volume ~nonpregnant; Na space ↓2 L
2 wk	Uterus 300 g ; placental site 3–4 cm; GFR normal; coitus may resume
3 wk	Rugae reappear; episiotomy healed; first fourth-trimester visit
4 wk	Uterus 100 g (involution complete)
4–6 wk	Vaginal epithelium proliferates; placental site extruded (6 wk); mature milk
6–8 wk	Menses return (non-breastfeeding); ovulation mean 7 wk ; ureters normal by 8 wk
12 wk	End of secondary PPH window and of VTE risk; comprehensive visit

Key Numbers to Memorize

- Puerperium **4–6 wk**; fourth trimester **12 wk** (visits at **3 wk** and **12 wk**)
- Uterus **1000** → **500** → **300** → **100 g** at 0, 1, 2, 4 wk; pregravid on US by **8 wk**
- Lochia rubra **3–4 d**, alba after **~10 d**; total **24–36 d**
- Endometrium restored by **day 16**; placental site **3–4 cm** at 2 wk, extruded by **6 wk**
- Secondary PPH **24 h – 12 wk**, **~1%**; methylergonovine **0.2 mg PO q3–4h × 24–48 h**
- WBC **≤25,000/μL**; VTE risk **12 wk**; CO normal by **10 d**
- Weight loss **5–6 kg** at delivery + **2–3 kg** diuresis
- Milk **≥600 mL/d**; whey:casein **60:40**; lactose **7 g/dL**; fat **4.2 g/dL**; protein **1.1 g/dL**
- Engorgement peak **3–5 d**; fever **37.8–39°C**, **<16 h**
- Retention: catheter **≥24 h**; residual **>200 mL** → recatheterize; bacteriuria **40%**
- Symphysis normal **0.4–0.5 cm**; **>1 cm** diastasis; surgery **>4 cm**
- Neuropathy **~1%** (LFCN 24%, femoral 14%)
- Anti-D **300 μg**; ovulation mean **7 wk** (as early as <28 d); breastfeeding pregnancy risk **~4%/yr**
- Hospital stay **48 h** vaginal / **96 h** cesarean